

Unit 6: Worked Examples

1. Find the prime factorization of each value. Use exponents where appropriate.

a. 48

$$\begin{array}{c}
 48 \\
 / \quad \backslash \\
 8 \quad 6 \\
 / \backslash / \backslash \\
 4 \quad 2 \quad 2 \quad 3 \\
 / \backslash \\
 2 \quad 2
 \end{array}$$

$2^4 \cdot 3$

c. 280

$$\begin{array}{c}
 280 \\
 / \quad \backslash \\
 28 \quad 10 \\
 / \backslash / \backslash \\
 4 \quad 7 \quad 5 \quad 2 \\
 / \backslash \\
 2 \quad 2
 \end{array}$$

$2^3 \cdot 5 \cdot 7$

c. 75

$$\begin{array}{c}
 75 \\
 / \quad \backslash \\
 3 \quad 25 \\
 \quad / \backslash \\
 \quad 5 \quad 5
 \end{array}$$

$5^2 \cdot 3$

2. Evaluate.

a. $14 + (121)$

$$\begin{array}{r}
 121 \\
 + 14 \\
 \hline
 135
 \end{array}$$

135

b. $-17 + 72$

$$\begin{array}{r}
 72 \\
 - 17 \\
 \hline
 55
 \end{array}$$

55

c. $-150 + (-140)$

$$\begin{array}{r}
 -150 \\
 + -140 \\
 \hline
 -290
 \end{array}$$

-290

d. $134 - (-156)$

$$\begin{array}{r}
 134 \\
 + 156 \\
 \hline
 290
 \end{array}$$

290

3. Fill out the table below.

-12×7 $ \begin{array}{r} -12 \\ \times 7 \\ \hline -84 \end{array} $ -84	$-4 \cdot (13)$ $ \begin{array}{r} 13 \\ \times -4 \\ \hline -52 \end{array} $ -52	$-200 \div 25$ $ \begin{array}{r} 8 \\ 25 \overline{) -200} \\ \underline{200} \\ 0 \end{array} $ -8	$\frac{-20}{-4}$ $ \begin{array}{r} 5 \\ -4 \overline{) -20} \end{array} $ negative negative = positive 5
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4. Evaluate each expression

a. $(-4)^3$

$$(-4)(-4)(-4)$$

$16(-4) = -64$

b. $(3)^3$

$$(3)(3)(3)$$

$9(3) = 27$

c. $(\frac{3}{4})^3$

$$(\frac{3}{4})(\frac{3}{4})(\frac{3}{4})$$

$(\frac{9}{16})(\frac{3}{4}) = \frac{27}{64}$

5. Tamara didn't realize she had a balance of \$19 in her bank account when she purchased a \$50 purse. When the purchase is processed, what will be her new account balance? Use the number line below to show your answer.

